

38.3 Sandwich covariance estimation

Theorem 38.2.2 left an asymmetry: the estimating equations are unbiased as soon as the mean model is right, while the identity that turns $\mathbf{A}_n = \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$ into a covariance needs the variance model too. When the latter fails, the covariance of $\hat{\boldsymbol{\beta}}$ becomes a product of three matrices instead of one, and the middle factor can be estimated from the residuals even though its n ingredients cannot.

38.3.1 The asymptotic distribution

Throughout $\boldsymbol{\beta}^0$ is the true parameter, $\eta_i^0 = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}^0$, $\mu_i^0 = h(\eta_i^0)$, and $\sigma_i^2 = \text{Var}(Y_i)$ is *not* assumed equal to $\phi V(\mu_i^0)/w_i$. Write

$$a(\eta) = \frac{h'(\eta)}{V\{h(\eta)\}}, \quad \mathbf{A}_n = \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{X} = \frac{1}{\phi} \sum_{i=1}^n w_i a(\eta_i^0) h'(\eta_i^0) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top,$$

and let $\mathbf{B}_n$ be the matrix Eq. 38.2.3. The conditions are

- (Q1) the Y_i are independent with $E Y_i = \mu_i^0$ for a $\boldsymbol{\beta}^0$ interior to the admissible set, p is fixed and $\text{rank}(\mathbf{X}) = p$;
- (Q2) $\|\mathbf{x}_{(i)}\| \leq C$ and $0 < w_{\min} \leq w_i \leq w_{\max} < \infty$;
- (Q3) on a neighbourhood $\mathcal{N}$ of $\boldsymbol{\beta}^0$, h and a are continuously differentiable with h' , a , a' bounded and h' , a bounded away from zero, uniformly in i and n ;
- (Q4) $\sigma_i^2 \geq \sigma_{\min}^2 > 0$, $E(Y_i - \mu_i^0)^4 \leq K$, and $n^{-1} \mathbf{A}_n \rightarrow \bar{\mathbf{A}}$, $n^{-1} \mathbf{B}_n \rightarrow \bar{\mathbf{B}}$ with both limits positive definite;
- (Q5) with probability tending to one Eq. 38.1.1 has a root $\hat{\boldsymbol{\beta}} \in \mathcal{N}$ with $\hat{\boldsymbol{\beta}} \rightarrow \boldsymbol{\beta}^0$ in probability.

Theorem 38.3.1 — Consistency, normality and the sandwich

Assume (Q1)–(Q5). Then:

(a)

$$\mathbf{B}_n^{-1/2} \mathbf{A}_n (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) \xrightarrow{d} N_p(\mathbf{0}, \mathbf{I}), \quad (38.3.1)$$

so that $\hat{\boldsymbol{\beta}}$ behaves as if its covariance were the **sandwich** $\mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1}$, which involves the distribution of $\mathbf{Y}$ only through its first two moments.

(b) If the variance specification Eq. 38.1.2 is correct, then $\mathbf{B}_n = \mathbf{A}_n$ by Theorem 38.2.2(d), and Eq. 38.3.1 becomes $\mathbf{A}_n^{1/2} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) \rightarrow N_p(\mathbf{0}, \mathbf{I})$: the **model-based** covariance $\mathbf{A}_n^{-1} = \phi (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ is correct, and by Proposition 38.2.3 no linear estimating function has a larger Godambe information — hence, whenever (Q1)–(Q5) hold for that estimating function too, none has a smaller asymptotic covariance.

(c) Let $\hat{c}_i = w_i h'(\hat{\eta}_i)/V(\hat{\mu}_i)$ and define the **empirical sandwich**

$$\hat{\Sigma}_{\text{rob}} = (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \left\{ \sum_{i=1}^n \hat{c}_i^2 (y_i - \hat{\mu}_i)^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right\} (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}, \quad (38.3.2)$$

in which ϕ cancels. Then $n \hat{\Sigma}_{\text{rob}} - n \mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1} \rightarrow \mathbf{0}$ in probability, and Eq. 38.3.1

holds with $\hat{\Sigma}_{\text{rob}}^{-1/2}(\hat{\beta} - \beta^0)$ in place of the left-hand side.

Proof. Step 1: the quasi-score is asymptotically normal. By Theorem 38.2.2(a) and (e), $\mathbf{U}_n(\beta^0) = \phi^{-1} \sum_i w_i a(\eta_i^0) \xi_i \mathbf{x}_{(i)}$ with $\xi_i = Y_i - \mu_i^0$ independent and mean zero and $\text{Cov}\{\mathbf{U}_n(\beta^0)\} = \mathbf{B}_n$. Fix a unit vector $\mathbf{v}$ and write $\mathbf{v}^\top \mathbf{B}_n^{-1/2} \mathbf{U}_n = \sum_i d_{ni} \xi_i$ with $d_{ni} = \phi^{-1} w_i a(\eta_i^0) \mathbf{v}^\top \mathbf{B}_n^{-1/2} \mathbf{x}_{(i)}$, so that $\sum_i d_{ni}^2 \sigma_i^2 = 1$. By (Q4), $\sigma_i^2 \leq K^{1/2}$ and hence $\sum_i d_{ni}^2 \geq K^{-1/2}$, while (Q2)–(Q4) give $d_{ni}^2 \leq \phi^{-2} w_{\max}^2 a_{\max}^2 C^2 \lambda_{\max}(\mathbf{B}_n^{-1}) = O(n^{-1})$, because $\lambda_{\min}(n^{-1} \mathbf{B}_n) \rightarrow \lambda_{\min}(\mathbf{B}) > 0$. So $\max_i d_{ni}^2 / \sum_i d_{ni}^2 \rightarrow 0$, which is Eq. 21.2.2, and the fourth moments are bounded: Lemma 21.2.1(b) with $\delta = 2$ and the Cramér–Wold device give $\mathbf{B}_n^{-1/2} \mathbf{U}_n(\beta^0) \rightarrow N_p(\mathbf{0}, \mathbf{I})$.

Step 2: the expansion. By (Q3), $\beta \mapsto \mathbf{U}_n(\beta)$ is continuously differentiable on $\mathcal{N}$, so on $\{\hat{\beta} \in \mathcal{N}\}$,

$$\mathbf{0} = \mathbf{U}_n(\hat{\beta}) = \mathbf{U}_n(\beta^0) + \bar{\mathbf{J}}_n (\hat{\beta} - \beta^0),$$

where

$$\bar{\mathbf{J}}_n = \int_0^1 \frac{\partial \mathbf{U}_n}{\partial \beta^\top} (\beta^0 + t(\hat{\beta} - \beta^0)) dt.$$

Step 3: the Jacobian. By the proof of Theorem 38.2.2(c),

$$\frac{\partial \mathbf{U}_n}{\partial \beta^\top}(\beta) = \frac{1}{\phi} \sum_{i=1}^n w_i \{a'(\eta_i) \{y_i - \mu_i(\beta)\} - a(\eta_i) h'(\eta_i)\} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top,$$

with $\eta_i = \mathbf{x}_{(i)}^\top \beta$; take $\|\beta - \beta^0\| \leq \delta$, so that every η_i moves by at most $C\delta$. The second term is nonrandom and equals $-\mathbf{A}_n$ at β^0 , and since ah' is uniformly continuous by (Q3) the two differ, after division by n , by at most $\phi^{-1} w_{\max} C^2 \omega(C\delta)$ with ω the modulus of continuity. In the first term write $y_i - \mu_i(\beta) = \xi_i + \{\mu_i^0 - \mu_i(\beta)\}$: the displacement is at most $h'_{\max} C\delta$, contributing $O(\delta)$ to n^{-1} times the norm, while $n^{-1} \phi^{-1} \sum_i w_i a'(\eta_i) \xi_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ has entries that average independent mean-zero variables of bounded variance, hence $o_p(1)$ by Chebyshev. By (Q5) δ may tend to zero, so $-n^{-1} \bar{\mathbf{J}}_n \rightarrow \bar{\mathbf{A}}$ in probability, with $\bar{\mathbf{A}}$ nonsingular.

Step 4: assembling. With probability tending to one $\bar{\mathbf{J}}_n$ is invertible and $\hat{\beta} - \beta^0 = (-\bar{\mathbf{J}}_n)^{-1} \mathbf{U}_n(\beta^0)$, so

$$\mathbf{B}_n^{-1/2} \mathbf{A}_n (\hat{\beta} - \beta^0) = \left(\mathbf{B}_n^{-1/2} \mathbf{A}_n (-\bar{\mathbf{J}}_n)^{-1} \mathbf{A}_n^{-1} \mathbf{B}_n^{1/2} \right) \mathbf{B}_n^{-1/2} \mathbf{U}_n(\beta^0).$$

By (Q4) and Step 3 the bracket tends in probability to $\mathbf{I}$, since $n^{-1} \mathbf{A}_n$, $n^{-1} \mathbf{B}_n$, their square roots and inverses all converge to fixed nonsingular limits. Slutsky and Step 1 give Eq. 38.3.1, and (b) is the substitution $\mathbf{B}_n = \mathbf{A}_n$.

Step 5: the empirical middle. Write $c_i = w_i a(\eta_i^0)$, so the middle of Eq. 38.3.2 at the true parameter is $\sum_i c_i^2 \xi_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ and $\phi^2 \mathbf{B}_n = \sum_i c_i^2 \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$. Each entry of $n^{-1} \sum_i c_i^2 (\xi_i^2 - \sigma_i^2) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ is an average of independent mean-zero variables whose variances are bounded by $c_{\max}^4 C^4 K$, so it is $o_p(1)$ by Chebyshev; this is Step 2 of the proof of Theorem 21.4.2. Replacing ξ_i by $y_i - \hat{\mu}_i$ and c_i by $\hat{c}_i$ costs a further $o_p(1)$, both displacements being bounded uniformly in i by a constant times $\|\hat{\beta} - \beta^0\|$, so Step 3 of that proof applies verbatim. The same reasoning gives $n^{-1} \mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X} \rightarrow \phi \bar{\mathbf{A}}$, and combining the three factors proves (c).

Remark (What (Q5) hides, and when it is free). Assuming a consistent root assumes something real: the equations can have no solution, or several. Under the canonical link of the variance function Q is strictly concave, and the argument for Theorem 34.5.1 gives existence and consistency without (Q5) as soon as $\lambda_{\min}(\mathbf{A}_n) \rightarrow \infty$; for a general link the conditions of Fahrmeir and Kaufmann (1985), quoted in Proposition 34.5.2, do the same and need only unbiasedness. Conditions (Q2)–(Q4) are the quasi-likelihood forms of the bounded-leverage and moment conditions of Theorem 21.4.2, and fail in the same places. ■

38.3.2 Model-based or robust

The two covariance estimates answer different questions. Under the log link with $V(\mu) = \mu$ and $w_i = 1$, where $\mathbf{W}_{ii} = \mu_i$ and $a(\eta_i) = 1$,

$$\mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1} = \left(\sum_i \mu_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right)^{-1} \left(\sum_i \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right) \left(\sum_i \mu_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right)^{-1},$$

against the model-based $\phi(\sum_i \mu_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top)^{-1}$. Only the *shape* of the dependence on μ matters, since a constant factor is absorbed into ϕ : a quasi-Poisson analysis is right about the standard errors whenever the true variance is proportional to the mean, whatever the constant, and wrong for any other shape.

Example. (Coverage when the variance function is right and when it is wrong)

A quasi-Poisson model $\log \mu_i = \beta_0 + \beta_1 x_i$ was fitted to data simulated with the same mean and two true variances, $\text{Var}(Y_i) = \mu_i + 2\mu_i^2$ (quadratic) and $\text{Var}(Y_i) = 9\mu_i$ (linear), over 3000 replicates at each of seven sample sizes. Figure 38.3.1 records the coverage of nominal 95% intervals for β_1 from four standard errors: Poisson with $\phi = 1$, model-based scaled by $\hat{\phi}$, the empirical sandwich Eq. 38.3.2, and a leverage-corrected sandwich described below.

Ignoring the dispersion is fatal in both panels and does not improve with n : 0.334 at $n = 20$ and 0.323 at $n = 1280$ in the quadratic case, 0.513 and 0.482 in the linear case. A linear true variance makes the specification correct and the $\hat{\phi}$ -scaled interval already good at $n = 20$, covering 0.919 against the sandwich's 0.880; a quadratic one stops it improving, at 0.914 for $n = 1280$, while the sandwich climbs to 0.941, buying correctness in the limit at the price of noise in small samples, as in Example (Coverage in a skewed design).

```
import numpy as np

BETA = np.array([1.5, 1.0])
HALF = 2.0 # the regressor runs over [-HALF, HALF]

def quasi_poisson(X, y, steps=25):
    """IRLS for the quasi-score equations with a log link and V(mu) = mu."""
    beta = np.zeros(X.shape[1])
```

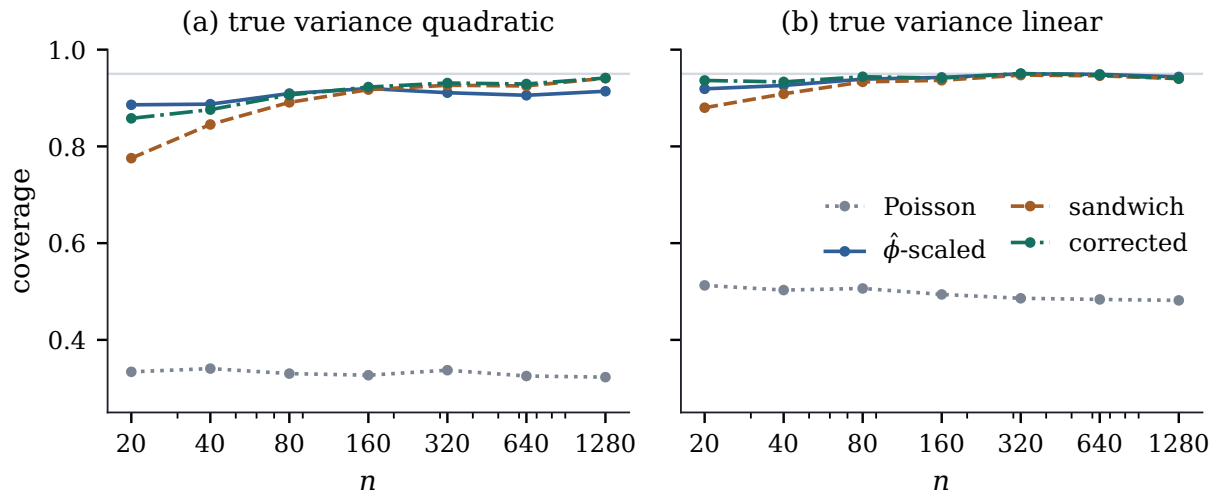


Figure 38.3.1: Coverage of nominal 95% intervals for the slope of a quasi-Poisson fit with a correct mean model. In (a) the true variance is quadratic in the mean, so the working variance function has the wrong shape; in (b) it is proportional to the mean.

```

beta[0] = np.log(max(y.mean(), 0.5))
for _ in range(steps):
    mu = np.exp(X @ beta)
    z = X @ beta + (y - mu) / mu
    beta = np.linalg.solve((X.T * mu) @ X, (X.T * mu) @ z)
return beta, np.exp(X @ beta)

def slope_variances(X, y):
    """The four estimates of Var(b1): naive, phi-scaled, sandwich, leverage-
    corrected."""
    beta, mu = quasi_poisson(X, y)
    n, p = X.shape
    bread = np.linalg.inv((X.T * mu) @ X)
    root = np.sqrt(mu)[:, None] * X
    hat = np.sum(root * (root @ bread), axis=1)
    r2 = (y - mu) ** 2
    phi = np.sum(r2 / mu) / (n - p)
    meat = lambda w: bread @ ((X * w[:, None]).T @ X) @ bread
    return beta[1], np.array([bread[1, 1], phi * bread[1, 1],
                             meat(r2)[1, 1], meat(r2 / (1 - hat) ** 2)[1, 1]])

def coverage(n, quadratic, reps, rng, kappa=0.5, phi=9.0):
    """Coverage of nominal 95% normal intervals for b1 over `reps` simulated data
    sets."""
    x = np.linspace(-HALF, HALF, n)
    X = np.column_stack([np.ones(n), x])
    mu = np.exp(X @ BETA)
    shape = np.full(n, kappa) if quadratic else mu / (phi - 1.0)
    hit = np.zeros(4)
    for _ in range(reps):
        y = rng.negative_binomial(shape, shape / (shape + mu)).astype(float)
        b1, var = slope_variances(X, y)
        hit += np.abs(b1 - BETA[1]) <= 1.96 * np.sqrt(var)
    return hit / reps

rng = np.random.default_rng(3821)
names = ["Poisson", "phi-scaled", "sandwich", "corrected"]

```

```

for n in (25, 100, 400):
    quad = coverage(n, True, 300, rng)
    lin = coverage(n, False, 300, rng)
    print(f"n = {n:4d}    quadratic variance " + " ".join(
        f"{k} {v:.3f}" for k, v in zip(names, quad)))
    print(f"                linear variance      " + " ".join(
        f"{k} {v:.3f}" for k, v in zip(names, lin)))

```

[Run this code online](#)

38.3.3 The dispersion, the quasi-deviance and tests

The model-based covariance needs ϕ , and so does any F test; neither $\hat{\beta}$ nor the sandwich Eq. 38.3.2 does. Since ϕ cancels from Eq. 38.1.1 and scales $\mathbf{A}_n$ by ϕ^{-1} , it is orthogonal to β : estimating it multiplies the asymptotic covariance of $\hat{\beta}$ by a consistent factor rather than inflating it.

Proposition 38.3.2 — Estimating the dispersion, and quasi-deviance tests

Assume Eq. 38.1.2 with the mean model correct.

- (a) $E[w_i(Y_i - \mu_i^0)^2/V(\mu_i^0)] = \phi$ for every i , so the **Pearson estimator** $\hat{\phi}_P = X^2/(n-p)$ with $X^2 = \sum_i w_i(y_i - \hat{\mu}_i)^2/V(\hat{\mu}_i)$ — the statistic Eq. 34.5.5 — is consistent for ϕ under (Q1)–(Q5). The divisor $n-p$ is an analogy with s^2 (Theorem 5.6.1), exact only in the normal linear model.
- (b) The **deviance estimator** $\hat{\phi}_D = D(\mathbf{y}, \hat{\mu})/(n-p)$, with D the quasi-deviance Eq. 38.2.2, is *not* in general consistent for ϕ . The two agree only so far as a third-order remainder is negligible: large binomial groups and large Poisson means, not small counts. Only $\hat{\phi}_P$ has expectation exactly ϕ at the true mean.
- (c) For nested mean models $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_0) = q$, fitted with the same V , the **quasi- F statistic** is

$$F = \frac{\{D(\mathbf{y}, \tilde{\mu}) - D(\mathbf{y}, \hat{\mu})\}/q}{\hat{\phi}_P}. \quad (38.3.3)$$

Under the reduced model and (Q1)–(Q5), $qF \rightarrow \chi^2(q)$, so referring F to $F(q, n-p)$ is asymptotically correct and is the usual small-sample convention. If the variance specification is wrong the limit fails and the Wald statistic built from Eq. 38.3.2 should be used instead.

- (d) No convention makes $-2Q(\hat{\beta})$ a log-likelihood, so Definition 29.2.2 does not apply. The criterion QAIC = $D(\mathbf{y}, \hat{\mu})/\hat{\phi} + 2p$, with $\hat{\phi}$ fixed at its value in the largest model, is in common use, but it is a convention, not a theorem. Cross-validation (Definition 29.3.1) needs no likelihood, and the analogue for correlated responses, the QIC of Chapter 40, rests on the same convention.

Proof.

- (a) Under Eq. 38.1.2, $E(Y_i - \mu_i^0)^2 = \phi V(\mu_i^0)/w_i$, which is the identity. The $T_i = w_i(Y_i - \mu_i^0)^2/V(\mu_i^0)$ are independent with mean ϕ and bounded variances, so

- $n^{-1} \sum_i T_i \rightarrow \phi$ by Chebyshev; replacing μ_i^0 by $\hat{\mu}_i$ costs $o_p(1)$, by Step 5 of the proof of [Theorem 38.3.1](#) with $\mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ replaced by 1, and $n/(n-p) \rightarrow 1$.
- (b) Each term of Eq. 38.2.2 is $2w_i \int_{\hat{\mu}_i}^{y_i} (y_i - t) V(t)^{-1} dt$. Expanding $1/V(t)$ about $t = y_i$ and integrating gives $w_i(y_i - \hat{\mu}_i)^2/V(y_i) + O(w_i|y_i - \hat{\mu}_i|^3)$, the calculation recorded after Eq. 34.5.5: a Pearson term with $V(y_i)$ in place of $V(\hat{\mu}_i)$. The two estimators therefore differ by the remainder and by the gap between those denominators, both negligible when $|y_i - \hat{\mu}_i|$ is small relative to μ_i and neither otherwise; there is no exact unbiasedness to appeal to, since $E d_i \neq \phi$ in general.
- (c) By Eq. 38.2.2 the difference is $2\phi\{Q(\hat{\beta}) - Q(\tilde{\beta})\}$. The proof of Proposition 34.5.3 used the log-likelihood only through the quadratic expansion Eq. 34.5.3, whose ingredients are asymptotic normality of the score and the identity between its variance and minus the expected Hessian. Under Eq. 38.1.2, Step 1 of the proof of [Theorem 38.3.1](#) supplies the first and Theorem 38.2.2(d) the second, with $\mathbf{A}_n$ in both roles, so that argument goes through word for word with Q in place of ℓ . Hence $\{D(\mathbf{y}, \tilde{\mu}) - D(\mathbf{y}, \hat{\mu})\}/\phi \rightarrow \chi^2(q)$; part (a) lets $\hat{\phi}_P$ replace ϕ , and $qF(q, n-p) \rightarrow \chi^2(q)$ makes the F reference asymptotically equivalent.
- (d) is a definition and a warning, not a claim.

■

Example. (Four standard errors for the strike data)

Continue Example (Strike durations). The Pearson statistic is 2393.5 on 60 degrees of freedom, so $\hat{\phi}_P = 39.89$, against $\hat{\phi}_D = 37.22$ from the quasi-deviance. The four standard errors for the production coefficient -7.680 are

covariance	standard error	t
Poisson, $\phi = 1$	0.406	-18.92
model-based, $\hat{\phi}_P$	2.564	-3.00
empirical sandwich	2.573	-2.98
leverage-corrected sandwich	2.896	-2.65

The Poisson standard error is too small by a factor of about six. More interesting are the second and third rows: the sandwich exceeds the model-based value by a factor of only 1.004, which is evidence *for* the quasi-Poisson variance function, so the model-based standard error is valid and slightly more efficient. The fourth row divides each squared residual by $(1 - h_{ii})^2$, with h_{ii} the leverages of the weighted hat matrix of Definition 34.5.2; the largest is 0.192, enough to matter at $n = 62$.

```
import numpy as np
import statsmodels.api as sm

data = sm.datasets.strikes.load_pandas().data
y = data["duration"].to_numpy(float)          # length of the strike, in days
x = data["iprod"].to_numpy(float)             # unanticipated industrial production
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape

def quasi_irls(X, y, V, steps=60):
```

```

"""Solve the quasi-score equations for a log link and variance function V."""
beta = np.zeros(X.shape[1])
beta[0] = np.log(y.mean())
for _ in range(steps):
    eta = X @ beta
    mu = np.exp(eta)
    W = mu ** 2 / V(mu) # working weight (dmu/deta)^2 / V(mu)
    z = eta + (y - mu) / mu # working response
    beta = np.linalg.solve((X.T * W) @ X, (X.T * W) @ z)
return beta

beta_lin = quasi_irls(X, y, lambda m: m) # V(mu) = mu
mu = np.exp(X @ beta_lin)
pearson = np.sum((y - mu) ** 2 / mu) # X^2 with V(mu) = mu
phi_pearson = pearson / (n - p)
W = mu # log link with V(mu) = mu: W_ii =
    mu_i
bread = np.linalg.inv((X.T * W) @ X) # (X' W X)^{-1}
root = np.sqrt(W)[: , None] * X
hat = np.sum(root * (root @ bread), axis=1) # leverages of the weighted hat matrix
resid = y - mu

cov_model = bread # phi = 1: the Poisson standard errors
cov_quasi = phi_pearson * bread # phi estimated from the same fit
cov_sand = bread @ ((X * resid[: , None] ** 2).T @ X) @ bread
cov_corr = bread @ ((X * (resid ** 2 / (1 - hat) ** 2)[: , None]).T @ X) @ bread

for name, C in [("Poisson", cov_model), ("quasi", cov_quasi),
                ("sandwich", cov_sand), ("corrected", cov_corr)]:
    se = np.sqrt(np.diag(C))
    print(f"{name:10s} slope {beta_lin[1]:8.3f} se {se[1]:6.3f} t {beta_lin[1] / se[1]:7.2f}")
print(f"largest leverage {hat.max():.4f}, sum {hat.sum():.4f}")

```

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38.3.4 Corrections, clusters and resampling

Three qualifications finish the section.

Small samples. The empirical sandwich is biased downwards, for the reason Proposition 21.4.1(a) identified in the linear model, and the remedy is the same: divide the i th squared residual by $1 - h_{ii}$ or by $(1 - h_{ii})^2$, with h_{ii} the diagonal of $\hat{\mathbf{W}}^{1/2} \mathbf{X} (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{W}}^{1/2}$, giving the analogues of HC2 and HC3 of Definition 21.4.1. The second is the “corrected” line of [Example \(Coverage when the variance function is right and when it is wrong\)](#), better than the plain sandwich at $n = 80$ in the quadratic panel, 0.906 against 0.891; Kauermann and Carroll (2001) give the leading bias term. The middle of [Eq. 38.3.2](#) is itself an average with very unequal weights, so it behaves like a variance estimate on far fewer than $n - p$ degrees of freedom (Example (One household carries the standard error)).

Clusters. If the observations come in G independent groups with arbitrary within-group de-

pendence, the middle of Eq. 38.3.2 becomes

$$\sum_{g=1}^G \left(\sum_{i \in g} \hat{c}_i (y_i - \hat{\mu}_i) \mathbf{x}_{(i)} \right) \left(\sum_{i \in g} \hat{c}_i (y_i - \hat{\mu}_i) \mathbf{x}_{(i)} \right)^\top,$$

the cluster-robust form Eq. 33.2.3 with the quasi-score's coefficients in place of the rows of $\mathbf{X}$. Its consistency needs $G \rightarrow \infty$, by Step 5 above with the cluster in place of the observation. The quasi-score is then no longer optimal, because the within-group correlations are ignored; Chapter 40 puts a generalized estimating equation in its place.

Resampling. The bootstrap of Chapter 23 applies unchanged: whole observations (Proposition 23.1.5), the wild bootstrap (Definition 23.2.1), or whole clusters (Exercise (C1)) for small G .

38.3.5 Exercises

38.3.5.1 A. Check your understanding

Exercise A1.

Under a log link and $V(\mu) = \mu$ with $w_i \equiv 1$, suppose the true variances are $\sigma_i^2 = \tau \mu_i$ for an unknown $\tau > 0$. Show that the sandwich and the model-based covariance have the same probability limit, and identify what $\hat{\phi}_P$ estimates. Repeat for $\sigma_i^2 = \tau \mu_i^2$ and show that the two then differ unless all μ_i are equal.

Exercise A2.

Verify from Eq. 38.3.2 that the empirical sandwich does not change if V is replaced by cV for a constant $c > 0$, and explain why this is to be expected.

38.3.5.2 B. Practice

Exercise B1.

Take an intercept-only quasi-Poisson model, $\log \mu_i = \beta$ for all i . Show that $\hat{\mu}_i = \bar{y}$, that the model-based variance of $\hat{\beta}$ is $\hat{\phi}_P / (n\bar{y})$, and that the sandwich variance is $\sum_i (y_i - \bar{y})^2 / (n\bar{y})^2$. Show that the two agree exactly when $\hat{\phi}_P$ is computed with divisor n rather than $n - p$.

Exercise B2.

Let $\text{Var}(Y_i) = \sigma_i^2$ and consider the quasi-score built from a working variance function V . Using Proposition 38.2.3 with the *true* variances, show that $\mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1}$ — the asymptotic covariance of Theorem 38.3.1(a) — is never smaller, in the nonnegative definite order, than the inverse Godambe information $(\sum_i h'(\eta_i)^2 \sigma_i^{-2} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top)^{-1}$ of the best linear estimating function, and that the two agree exactly when $\sigma_i^2 \propto V(\mu_i)/w_i$.

Solution. Repeat the proof of Proposition 38.2.3 with $\Sigma = \text{diag}(\sigma_i^2)$ in place of $\text{diag}\{\phi V(\mu_i)/w_i\}$; only its last display used the specification. The quasi-score built from V is a linear estimating function with $\mathbf{a}_i = w_i h'(\eta_i) \mathbf{x}_{(i)} / \{\phi V(\mu_i)\}$, so by Theorem 38.2.2(c),(e) its Godambe information is $\mathbf{A}_n \mathbf{B}_n^{-1} \mathbf{A}_n$, at most the maximum $\mathbf{K}^\top \mathbf{K} = \sum_i h'(\eta_i)^2 \sigma_i^{-2} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$; inverting reverses the order. Equality needs coefficients proportional to $h'(\eta_i) \mathbf{x}_{(i)} / \sigma_i^2$, that is $w_i / V(\mu_i) \propto 1 / \sigma_i^2$.

38.3.5.3 C. Going deeper

Exercise C1.

A large discrepancy between $\hat{\phi}_P(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$ and Eq. 38.3.2 is evidence that the variance specification is wrong. Propose a test statistic from the difference of the two matrices, say $\text{tr}\{(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X}) \hat{\Sigma}_{\text{rob}}\} - p \hat{\phi}_P$, argue informally that its mean is about zero when the specification is correct, and say what stands in the way of a usable null distribution. (This is White's information matrix test; compare Exercise (B3).)